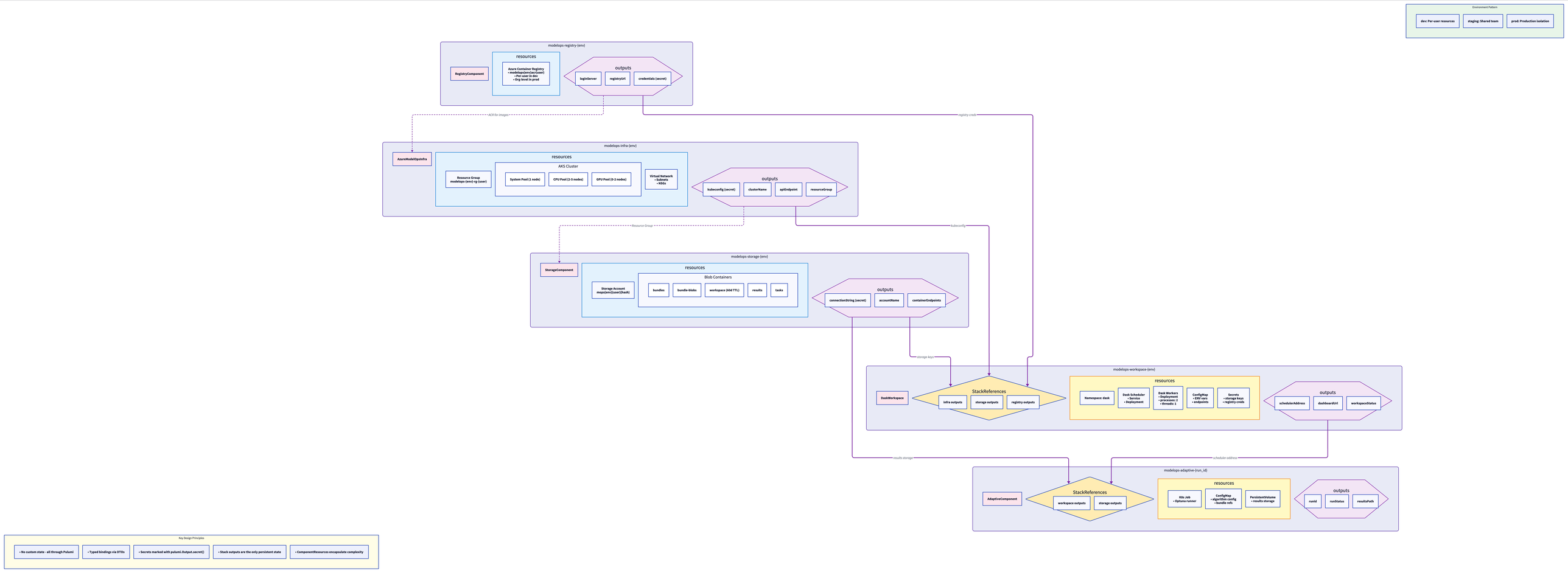
Environment Pattern
dev: Per-user resources
staging: Shared team
prod: Production isolation
modelops-registry-{env}
RegistryComponent
resources
Azure Container Registry
• modelopsregistry{user}
• Per-user in dev
• Org-level in prod
outputs
loginServer
registryUrl
credentials (secret)
ACR for images
registry creds
modelops-infra-{env}
AzureModelOpsInfra
resources
Resource Group
modelops-{env}-rg-{user}
AKS Cluster
System Pool (1 node)
CPU Pool (2-3 nodes)
GPU Pool (0-2 nodes)
Virtual Network
• Subnets
• NSGs
outputs
kubeconfig (secret)
clusterName
apiEndpoint
resourceGroup
Resource Group
kubeconfig
modelops-storage-{env}
StorageComponent
resources
Storage Account
mopo{env}{user}{hash}
Blob Containers
bundles
bundle-blobs
workspace (60d TTL)
results
tasks
outputs
connectionString (secret)
accountName
containerEndpoints
storage keys
modelops-workspace-{env}
DaskWorkspace
StackReferences
infra outputs
storage outputs
registry outputs
resources
Namespace: dask
Dask Scheduler
• Service
• Deployment
Dask Workers
• Deployment
• processes: 2
• threads: 1
ConfigMap
• ENV vars
• endpoints
Secrets
• storage keys
• registry creds
outputs
schedulerAddress
dashboardUrl
workspaceStatus
results storage
scheduler address
modelops-adaptive-{run_id}
AdaptiveComponent
StackReferences
workspace outputs
storage outputs
resources
K8s Job
• Optuna runner
ConfigMap
• algorithm config
• bundle refs
PersistentVolume
• results storage
outputs
runId
runStatus
resultsPath
Key Design Principles
• No custom state - all through Pulumi
• Typed bindings via DTOs
• Secrets marked with pulumi.Output.secret()
• Stack outputs are the only persistent state
• ComponentResources encapsulate complexity